

Engineering Report

Triangle Park Ithaca Flood Control Proposal

BEE 4730: Watershed Engineering

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Executive Summary

This project proposes a conceptual redesign of Triangle Park, also known as Thompson Park, in Ithaca, NY, to function as a temporary flood detention area for Cascadilla Creek. The core objective is to provide a viable watershed engineering solution to mitigate the potential for extensive chronic flooding in the downtown Ithaca area.

The redesign and analysis focus on the following key components,

- Design Storm: The solution is modeled based on a 100-year 24-hour design storm event, which provides the critical parameters for determining the maximum flow and volume of stormwater runoff from the Cascadilla Creek Watershed.
    - Maximum Runoff Flow: The analysis based on a composite hydrograph, determined the maximum flow rate for this storm is 47.98 m^3/s or 1694 cfs
    - Total Runoff Volume: The total volume of runoff for the entire Cascadilla Creek Watershed from the design storm is approximately 41,400,000 m^3 or 146,000,000 ft^3.
- Park Redesign: The proposed solution involves adjusting the park's topography to create a detention basin by lowering the elevation of the park's lower surface. The park is a small, highly urbanized green space adjacent to the concrete-lined reach of Cascadilla Creek.

The conceptual redesign involves modifying the park's edge along the creek, which currently sits 2.5 ft above the stream bed, by lowering the main park level to1 ft above the stream bed. A 6 ft long, 1.5 ft high embankment will connect the new lower surface park to the surrounding sidewalk. The park covers an area of approximately 55,000 ft^2. By lowering the park to this level it would be able to hold an additional 85,700 ft^3 of water storage capacity during high flow events.

The redesign is anticipated to provide reliable protection from floods caused by storms, offering increasing importance as such extreme storms may become more frequent due to changing climate conditions. However, the conversion to a temporary flood retention basin will change the park's role as a recreational space, though it should largely be able to serve its current purpose except during and immediately after an extreme weather event.

Scope and Justification

The project's scope is to reconceptualize Triangle Park–also referred to as Thompson Park–to function as a temporary flood detention area for Cascadilla Creek. The conceptual redesign and scope of the project includes the following:

- Hydrological Analysis and Design Parameters: Utilizing a 100-year 24-hour design storm for the creation of a hydrograph to determine the maximum flow and volume of stormwater runoff from the Cascadilla Creek Watershed.
- Conceptual Redesign and Capacity Assessment: Adjusting the park's topography to create a detention basin by lowering the elevation of the park's lower surface elevation to increase its stormwater capacity.

The justification for this project is to provide a viable watershed engineering solution to mitigate the potential for extensive chronic flooding in the City of Ithaca, focusing on the highly susceptible downtown area. The project works to address potential flood risks through the conversion of an existing urban green space to additionally work as a piece of critical flood infrastructure for the Cascadilla Creek watershed.

Site Description

Triangle Park, also known as Thompson Park, is a small, highly urbanized green space located along Cascadillia Creek in downtown Ithaca, New York. The park occupies a triangular shaped parcel immediately adjacent to the concrete lined reach of Cascadillia Creek that runs through the city of Ithaca. Aerial measurements from Google Maps indicate that Triangle Park covers an area of approximately 57,000 ft^2 or about 1.25 acres. The long axis of the park is oriented parallel to the creek, with the triangle's based along the upstream end of the park and the apex pointing downstream. The park slopes from the surrounding streets toward the top of the creek's retaining wall which functions as a narrow strip of open space between the roadway network and the flood control channel.

Cascadilla Creek next to the park flows in the concrete channel. The channel is around 29 feet wide, with straight concrete walls and a flat rocky bottom. The concrete walls one side has a height of 3 ft 10 inches while the other side of the wall is around 2.5 ft tall. These heights were measured from the channel bottom up to the top of each wall. The concrete wall along the park continues for about 180 meters on both sides of the creek. The channel is designed to move water quickly through the city and not store it or provide much habitat. It was estimated that this reach of the creek can carry a maximum flow of about 606.7 cfs (17.18 m^3/s) at an average velocity of about 8.37 ft/s (2.55 m/s) before overflowing.

Within Triangle Park, land cover is mainly dominated by a maintained lawn with scattered trees and limited vegetation. The surface is open and unobstructed with existing amenities such as walk paths and benches located near the higher perimeter of the park adjacent to the sidewalk, above the retaining wall. Soils at this site are typical of hydrological soil group C, which consists of loam to clay loam textures with relatively low to moderate infiltration capacity. These soils are able to provide some infiltration and temporary storage, but not as permeable as sandy soils and thus more prone to generate runoff during larger storm events.

Overall, Triangle Park is located in an area with open space adjacent to an engineered flood control channel. With an area of around 57,000 ft^2 it gently has a sloped topography towards the creek. Along with moderately permeable soils, this makes Triangle Park a promising candidate for regrading into a temporary flood detention storage area to help mitigate extreme flows along Cascadilla Creek while maintaining its ability to be an accessible park.

Watershed Characteristics

Watershed

The Cascadilla Creek Watershed encompasses an area of about 13.5 square miles (35 square kilometers), as calculated using StreamStats, a USGS-based software. Cascadilla Creek is 12 miles long, and runs through Cornell University's campus towards its outlet at Cayuga Lake Inlet in Downtown Ithaca, New York. Cascadilla Creek's flow has been reengineered to follow straight concrete channels in Downtown Ithaca, including the part of the Creek that runs adjacent to Triangle Park.

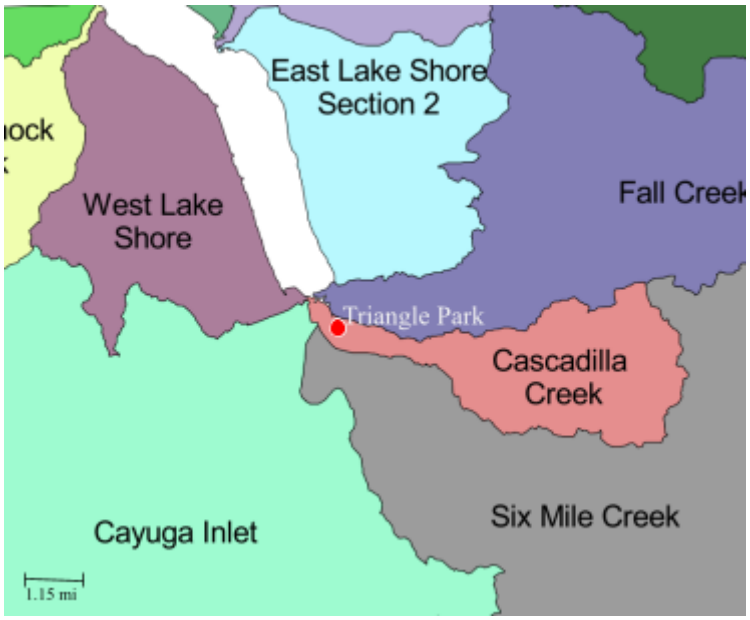


Figure 1. Location of Triangle Park within the Cascadilla Creek Watershed (adapted from TC Stormwater)

Climate

The Ithaca and Central New York region has a four-season climate. The region is considered part of the Finger Lakes region within Central New York and is susceptible to large rain events and snow events as well. The winter season lasts for about five months from November to March with spring and fall being about two months long each. Summer is warm and sunny and lasts from June to August. The 100-year 24-hour flood event that is the basis for modeling this report's engineering solution is calculated based on

the Northeast Regional Climate Center's extreme precipitation data as discussed further in Appendix A, under the Design Storm Characteristics subsection. As the effects of climate change further impact the Ithaca climate, flooding can also become more difficult to predict. In the past few years, droughts continue to extend for longer periods of time, as climate change heralds in new patterns not experienced in the region for long periods of time. The droughts cause soil to become completely unsaturated and when rain does fall, erosion and associated flooding become more problematic. Storms have also become more intensive over the past decade, another indication of climate change's effects. Therefore, when large storms follow elongated periods of drought, the soil does not have the capacity to hold the water, and erosion and runoff cause more flooding.

Runoff Calculation

The design storm chosen for the park redesign is the 100-yr 24-hr storm, which represents a very extreme precipitation situation. The Curve Number approach was used along with the time of concentration to create hydrographs that predict runoff rate and volume, which were used to assess the flood mitigation potential of the park redesign. The graph in Figure 2 displays the composite hydrograph of this storm that was used to get the maximum flow rate of the runoff as the main constraint in designing a flood mitigation solution. The maximum flow rate is 1,694 cfs or 47.98 m^3/s. More details and related calculations can be found in Appendix A.

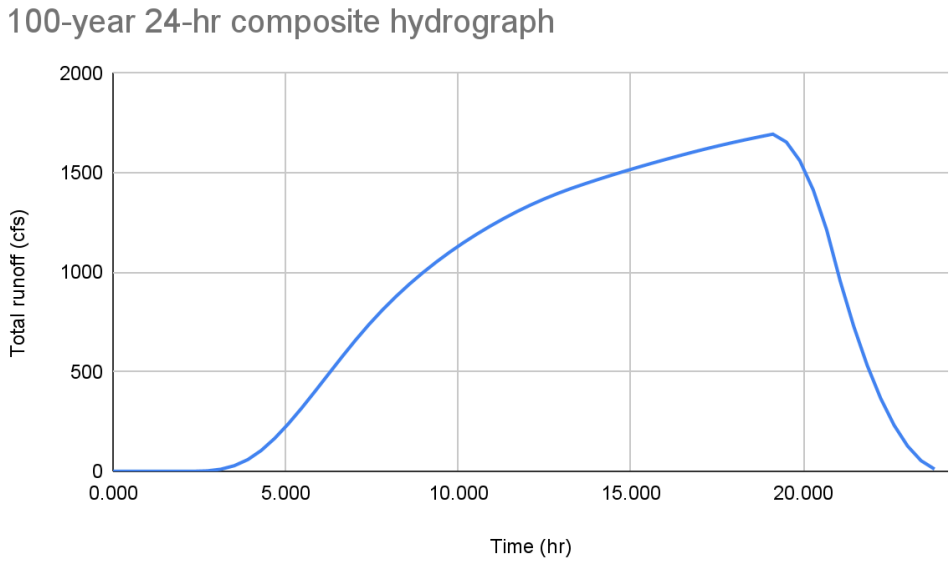


Figure 2. Graph of the 100-year 24-hour composite hydrograph.

Proposed Design

Description of Park Redesign

The park is a triangular shape with the steam running along the longest side. At this side, the stream sits in a bed approximately 29 ft wide with a stony bed and the sides lined by concrete. The stream edge on the

park's side sits lower, 2.5 ft high at its lowest, with the rest of the park sitting level with the upper edge of this wall. The park itself is 242 ft and 472 ft long, for an area of approximately 57,000 ft^2.

To convert the park to a flood control system, the changes in Figure 3 are proposed. This constitutes lowering the wall and park height from 2.5 ft to 1 ft above the stream bed at its lowest, with an embankment 6 feet long and rising 1.5 ft up to the sidewalk.

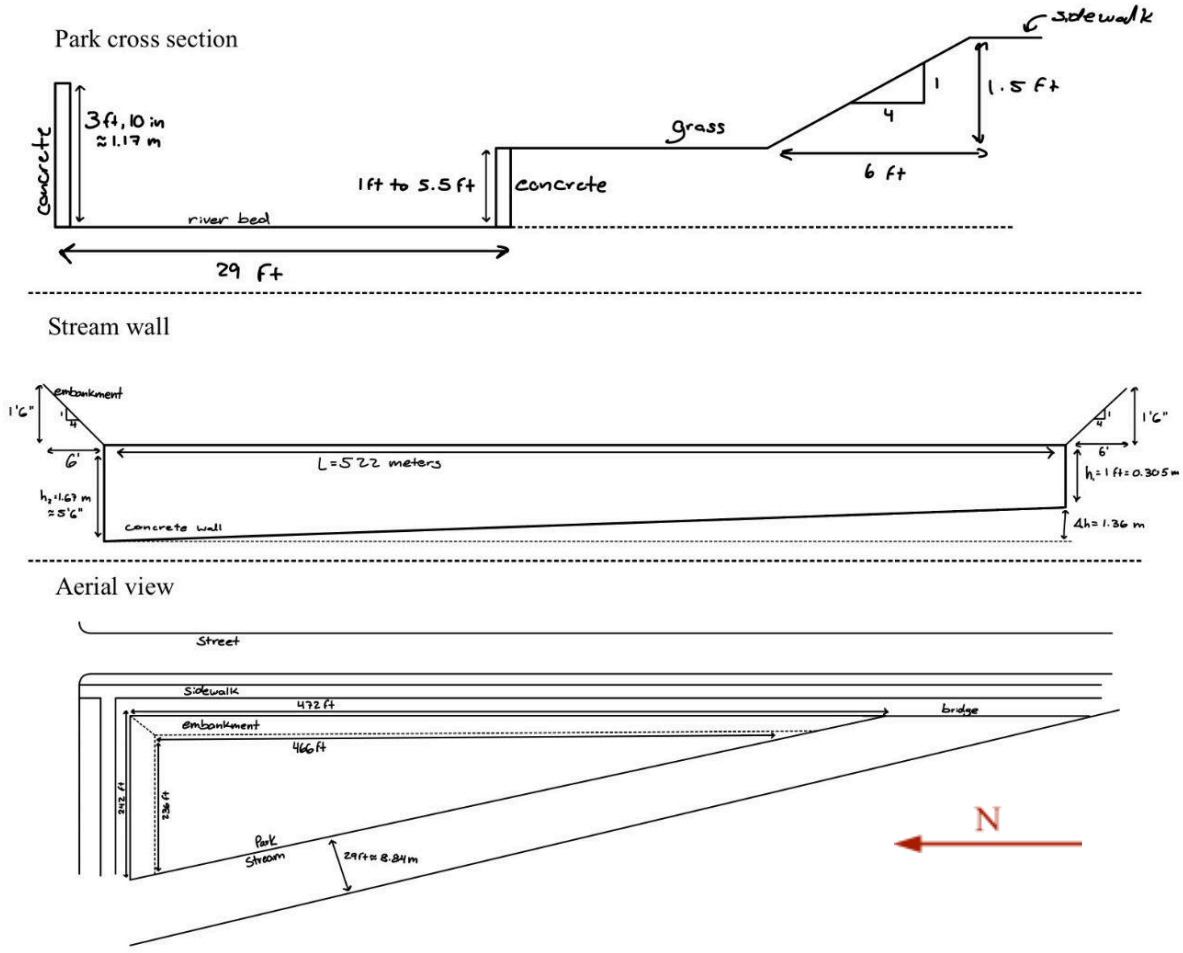


Figure 3. The design cross-section, side-view, and top-view to be applied as a flood mitigation strategy at Triangle Park.

This design takes advantage of the park's position next to the river to act as flood plain when flows in the stream exceed a height of 1 ft, the lowest height of the wall. While down the rest of the length the park shares with the stream the walls are higher than this, when flow exceeds 1 foot of height it will overflow into the park, preventing the height from increasing further. The total height difference between the streets and the stream bed remains the same, meaning the park will fill in its entirety before overflowing into the street. However, the frequency that the stream would overflow both the stream channel and the park is uncertain. From calculations performed with the previous lower wall, which was 2.5 ft high, in Appendix B, it is safe to assume that flooding over the other, 3 ft 10 inch high wall would occur much less frequently than flooding into the park. Furthermore, the park redesign would reduce the frequency of

which flooding occurs over the other wall because of its ability to capture and distribute any overflow from the channel. Further study of flooding from the channel is included in the next section, and may require future work.

The park additionally features an embankment from the park to the sidewalk for continuity and to maintain access to the park for recreational use during times when flood risk is low. The embankment was designed to have a 1:4 slope to allow for mowing requirements to be met, and also leaving open the possibility of installation of stairs for ease of access. This design therefore provides flood mitigation capacity while maintaining the park's recreational potential, as area of the park is only minorly reduced. There will be plantings in the park along the channel to create a vegetated buffer that both reduces runoff into the stream from the park and provides aesthetic value.

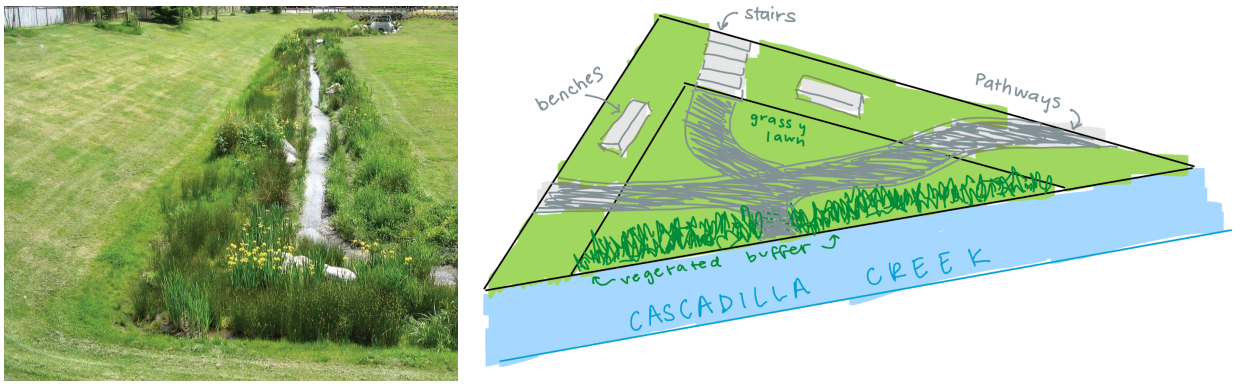


Figure 4. Examples of buffered lawns and an illustration of the final design of Triangle Park with recreational and aesthetic features included.

Erosion concerns are most likely minimal, although still considered. Because the park redesign does not alter stream flow path and only affects a stretch of approximately 1,700 ft, erosion should not be significantly affected for the stream bed itself. The stream walls are made of concrete, which should be able to resist erosion fairly well. One potential issue arises with the drainage of the park itself. As water overflows the streams during high flow periods, it is likely stream debris and sediment may be washed into the park itself, left behind after the high flow event is over. This presents the possibility of debris buildup in the park, which would require manual removal, or the buildup of sediment in the park. There is a low risk of erosion on the park embankment, as water filling up the park will flow at low speeds, minimizing any erosion risk. Since the park sits level with the stream wall, surface water runoff into the stream most likely would cause minimal erosion due to low flow speeds, but it is recommended to include a vegetation buffer, particularly next to the stream, to prevent soil erosion. This same strategy can be used on the park embankment to help minimize any erosion risk and for aesthetic purposes.

Flood Mitigation Capacity

The park redesign allows for Triangle Park to be used as a temporary detention basin when Cascadilla Creek overflows. With the increase of the park's volume, an additional 85,700 ft^3 of water can be stored during high flow events (see Appendix B for calculations). This is approximately 0.06% of the total runoff volume for the entire watershed from the 24-hr 100-yr design storm. However, scaling the total

runoff volume to the area of the park itself, the added volume can retain approximately 37.4% of total runoff from the park itself during such a large storm. This suggests that the new design for the park is able to reduce the amount of water entering Cascadilla Creek during big storms, and will also be able to temper the flow of smaller, more-frequent storms as well.

Upon assessing the runoff volume for storms with shorter durations and increased frequency, the park was found to detain a slightly larger amount of water equivalent to the 6-hr 2-yr storm for this region (see Appendix B for calculations). This storm generates approximately 79,600 ft^3 of runoff, which is less than the total volume of the park, which is 85,700 ft^3. This can considerably reduce the water flow through the Cascadilla Creek channel and detain excess runoff during larger storm events and prevent flooding from impacting the streets and neighborhoods surrounding the creek.

Additionally, with the slight decrease in channel height on one side, the maximum flow that the channel can handle is lower than the existing design. The existing design handles a maximum channel flow of 606.7 cfs, and there is 50% of a storm that can cause flooding annually. The new design handles 145 cfs – about 24% of the existing channel. However, this maximum flow is still considerably high, and storms that generate a comparable amount of runoff as a 12-hr 1-yr storm can still be contained by the new channel. This suggests that storms of shorter duration are not likely to overflow the channel and flood the park area. The channel will begin to overflow for most 5-yr storms, which have maximum flow rates slightly above 200 cfs, and some longer duration 2-yr storms. This is comparable to the current channel conditions, so flood risk is not increased by alterations to the lower channel wall. Thus, the park is expected to flood every few years or less and only for a short period of time, which allows for it to remain a usable recreational space for a majority of the time.

Future Work

Caveats

The ability of this redesign to limit flooding in Ithaca is contingent on the magnitude of the storm that it experiences. The redesign considers a 24-hr-100-year storm event when considering the magnitude of a storm and associated flood. However, this figure results from historical data. Due to uncertainty in climatic conditions of the future, these storms may increase in magnitude or frequency. The 24-hr-100-year storm event of the next century will likely differ from that of the past century, and may render this redesign less effective than currently anticipated. However, the redesign is based on floods that are caused by storms of this size. As these storms become more frequent in the future, the protection offered by the redesign of Triangle Park becomes increasingly important for mitigating the impact of floods in the Ithaca area.

The Triangle Park redesign is meant to hold a small percentage of the volume of the Cascadilla Creek Watershed. Because the Triangle Park redesign is limited by the size and location of the park, it is unable to offer complete, comprehensive protection against a 24-hour-100-year magnitude flood. However, the aim of this redesign is to address flooding on a local, neighborhood level, preventing local residents and homeowners from suffering from the effects of a flood. Such a redesign will be useful for more frequent,

smaller magnitude storm events and flash floods, as well as comprising part of a broader flood management plan for the watershed.

This redesign will change the role of Triangle Park as a recreational space. Currently, Triangle Park is open to the public during all seasons under all types of weather. As a green space located in the center of the urbanized area of Ithaca, Triangle Park serves as an important source of recreation and outdoor access for residents of and visitors to the city. This redesign of Triangle Park is designed to maintain Triangle Park's recreational capability as much as possible while also serving as a location for an intervention to reduce the impact of floods from the Cascadilla Creek Watershed. Because this redesign functions as a temporary flood retention basin, rather than a permanently filled pond or reservoir, Triangle Park should largely be able to serve its current purpose except for during times of extreme weather. When the temporary retention basin is not filled with water, visitors to the park will be able to enjoy its rich recreational offerings. During times of flooding, the park will not be suitable for recreation as large areas of it will be submerged under flowing water, which may be unsafe for visitors. After the park drains following a storm event, there may be debris remaining in the previously submerged basin, which may need to be cleared out before the park is safe for recreation. This would involve temporary cleaning or park maintenance crews hired by the municipality.

Future Work

This report provides an initial analysis of one potential design to mitigate potential 24-hr-100-year floods in Ithaca at Triangle Park. A simple solution was considered in this project, and other designs were not considered. Therefore, future work will require an in-depth comparison of the various methods proposed to mitigate flooding via the re-engineering of Triangle Park. By comparing the cost, feasibility, environmental impact, and usability of the park, a more holistic-based decision can be made as to the best option to decrease flood risk for the downtown Ithaca area.

Furthermore, future work will require more field work at the Triangle Park location to fully determine the feasibility of the proposed solution. Initially, only the measurements of Cascadilla Creek were taken during field work, and more will have to be conducted and analyzed to ensure that this proposal is fully feasible. Overall, this proposal provides a framework and solution to the potential risk of flooding in downtown Ithaca but future work is required to increase the reliability of the results so that the solution can be confidently implemented.

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Appendices

Appendix A

Calculations performed to determine the maximum flow of the design storm are included here in Appendix A.

Design Storm Characteristics

The 24-hour 100-year storm was chosen as the design storm for this project. From the NRCC extreme precipitation data for Ithaca, New York, the average precipitation for the design storm was found to be 5.92 inches, or approximately 150 mm.

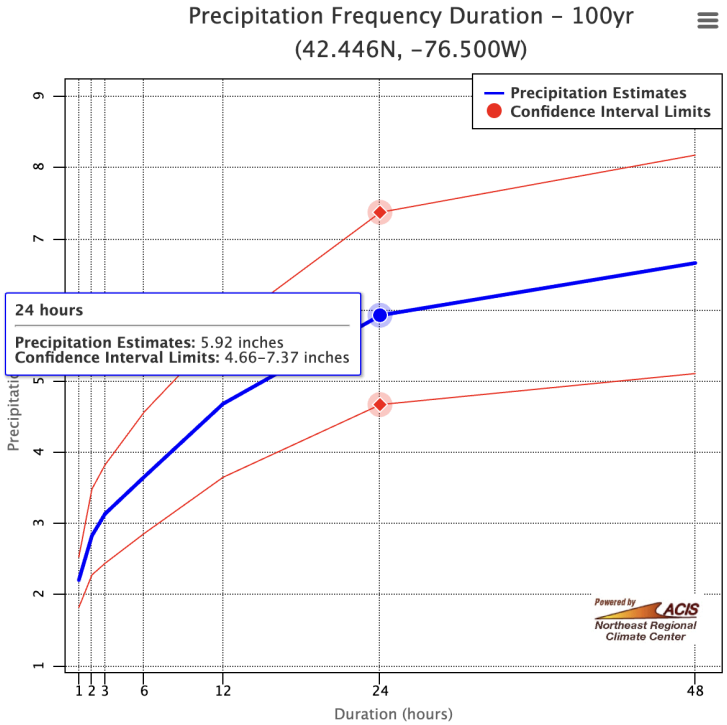


Figure A1. NRCC extreme precipitation graph for the 24-hr 100-yr storm in Ithaca, New York.

Then, the Kirpich method was used to determine the time of concentration (t_c) for the Cascadilla Creek watershed. Given the following watershed characteristics, t_c can be calculated with Equation A1:

$$t_c = 0.0078L^{0.77}S^{-0.385} \qquad (1)$$

$$S = (elevation_{highest} - elevation_{outlet})/L$$

- Watershed area ≈ 35 km^2
- Elevation at highest point in the watershed ≈ 1,870 ft amsl
- Elevation at watershed outlet ≈ 380 feet amsl
- Longest flow path (L) ≈ 67,000 ft

Using these values for the Cascadilla Creek watershed, t_c is found to be 176 min.

Next, the curve number (CN) can be calculated from the given watershed land use.

- 40% forested, good condition woods
- 23% row crops, good condition straight row crops
- 20% hay and pasture, good condition
- 15% residential, ⅓ acre
- 2% paved

The equation to calculate CN is shown in Equation A2. Intermediate values can be found from the soil type B columns from the Tables 2-2 from the American Society of Civil Engineers (ASCE) Curve Number Hydrology book.

$$CN = 0.4(forested) + 0.23(crops) + 0.2(pasture) + 0.15(residential) + 0.02(paved)$$
$$CN = 0.4(70) + 0.23(85) + 0.2(74) + 0.15(90) + 0.02(98) = 77.81 \quad (2)$$

Composite Unit Hydrograph

To create the unit hydrograph, the slope (S) and initial abstraction (Ia) were first calculated. Equations A3 and A4 show the calculations.

$$S = 1000/CN - 10 = 2.85\ inch\ or\ 72.43\ mm \quad (3)$$
$$I_a = 0.2S = 0.57\ inch\ or\ 14.49\ mm \quad (4)$$

Additionally, the time to peak runoff (t_p) and the time of recession (t_r) are determined from t_c from the following equations, where delta is 0.133 times t_c:

$$t_p = \Delta/2 + 0.6 * t_c = 1.96\ hr \quad (5)$$
$$t_r = 1.67 * t_p = 3.26\ hr \quad (6)$$

Next, the runoff volume and peak runoff rate were calculated.

$$Q_v = A * 1mm = 35m^2 * 0.0001m = 35000\ m^3 \quad (7)$$

$$q_p = (2 * Q_v)/(t_p + t_r) = 3.72\ m^3/s \quad (8)$$

The calculations used to determine the runoff rate are summarized in Figure A2.

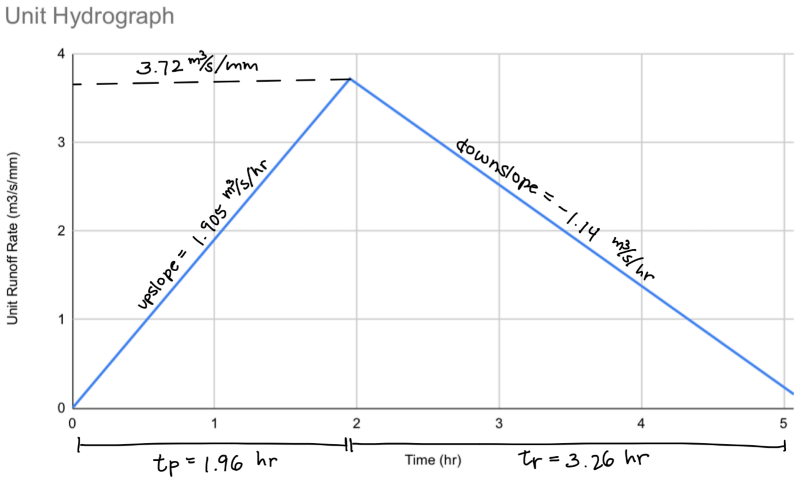


Figure A2. Unit hydrograph with peak runoff rate, slopes, and time to peak and recession labeled.

The slopes of the unit hydrograph are used to calculate the runoff rate from the start of the storm until it has receded. Table A1 shows the first few values for the time, runoff rate, precipitation, and volumetric runoff flow rate as an example. The full table can be found in the supplemental materials section.

Table A1. Sample values used to calculate the composite hydrograph. A full table of values can be found in the supplemental materials section after the appendices.

time (hr)	U (mm/hr)	P (mm)	Q (m3/s)	deltaQ (m3/s)
0.000	0.000	0.000	0.000	0.000
0.390	0.743	2.425	0.000	0.000
0.780	1.487	4.851	0.000	0.000
1.170	2.230	7.276	0.000	0.000
1.561	2.973	9.701	0.000	0.000
1.951	3.717	12.126	0.000	0.000
2.341	3.271	14.552	0.000	0.000
2.731	2.826	16.977	0.083	0.083
3.121	2.381	19.402	0.312	0.230
3.511	1.936	21.828	0.675	0.363
3.901	1.491	24.253	1.160	0.485
4.291	1.046	26.678	1.756	0.596
4.682	0.601	29.103	2.454	0.698
5.072	0.156	31.529	3.246	0.792

The unit hydrographs for the 24-hr 100-yr design storm are illustrated in the figure below. The composite hydrograph, which is the sum of all the unit hydrographs at a given time, is in the main body report.

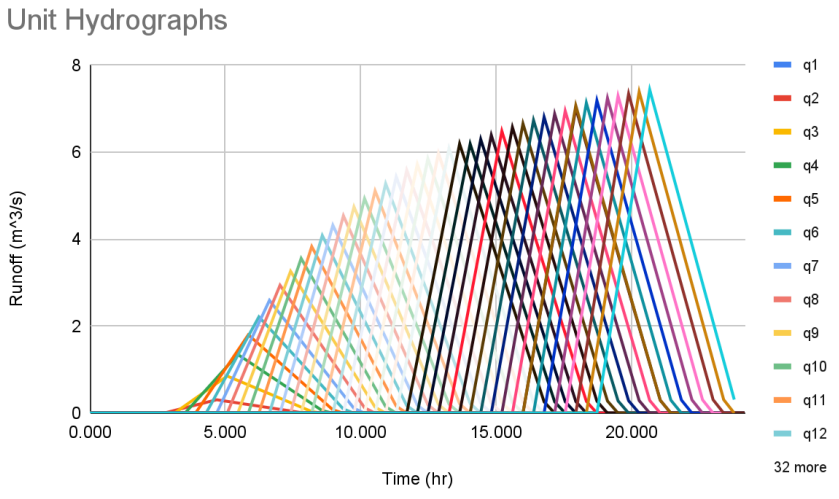


Figure A3. Graph plotting all the unit hydrographs from the start of the storm through its 24-hr duration.

The maximum flow rate, as mentioned in the main body report, is 47.98 m^3/s or 1694.26 cfs. The total volume of runoff for the entire Cascadilla Creek Watershed was calculated to be 41,400,000 m^3 or 146,000,000 ft^3.

Appendix B

The feasibility of the proposed design to detain an amount of the volume of stormwater (calculations shown in Appendix A) is supported by calculations included in Appendix B.

The dimensions of the proposed park design are shown in the figure below. Area and perimeter measurements were taken using ArcGIS Online MapViewer. The length of the base is 241.69 ft, and the length of the height is 471.82 ft. The total area of the new Triangle Park. minus the embankments, is approximately 55,000 ft^2.

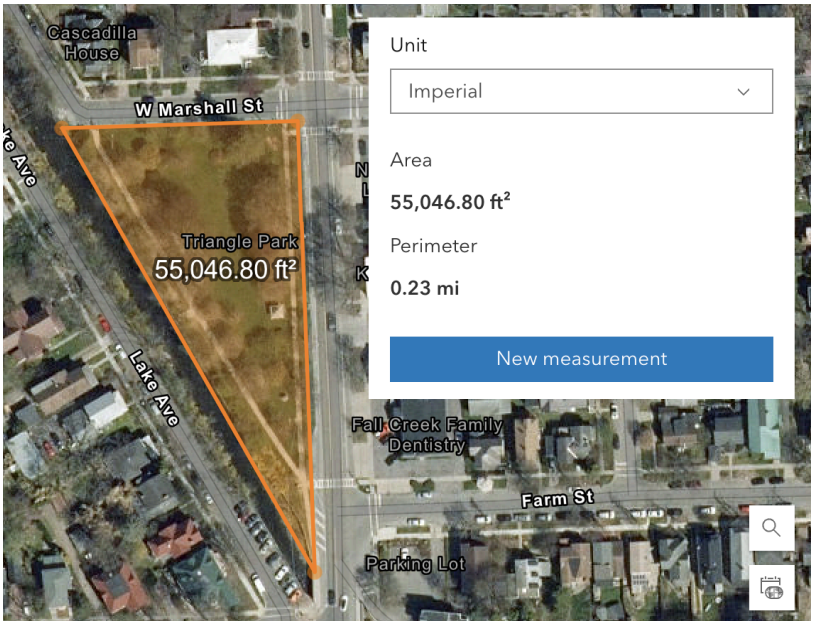


Figure B1. Aerial image of Triangle Park taken from ArcGIS Online MapViewer, with the area labeled.

Since the depth of the park design beneath the sidewalk elevation is 1.5 ft, the height of water reached by the maximum volume of stormwater must be less than 1.5 ft. A diagram of the park is shown below in Figure B2.

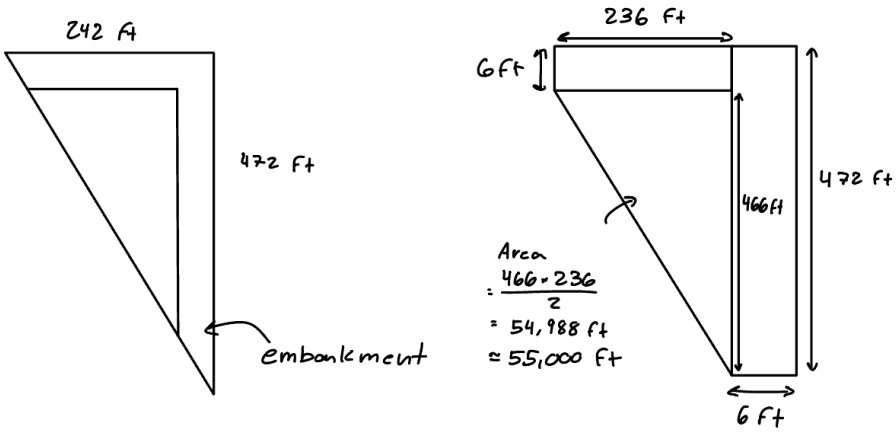


Figure B2. Diagram of the Triangle Park redesign showing the sloped embankment and lower surface.

The total volume of water that the new park design can detain is 85,700 ft^3.

The equivalent storm for this volume of water was found by reducing the storm duration and increasing storm frequency. Using the same hydrograph methods described in Appendix A, the precipitation that produces a runoff volume approximately equal to or less than the new park design was found to be 1.57 inch over six hours, which has a total runoff volume of 79600 ft^3.

This corresponds to the 6-hr 2-yr storm for this region. Runoff does not occur until about three and a half hours into the storm.

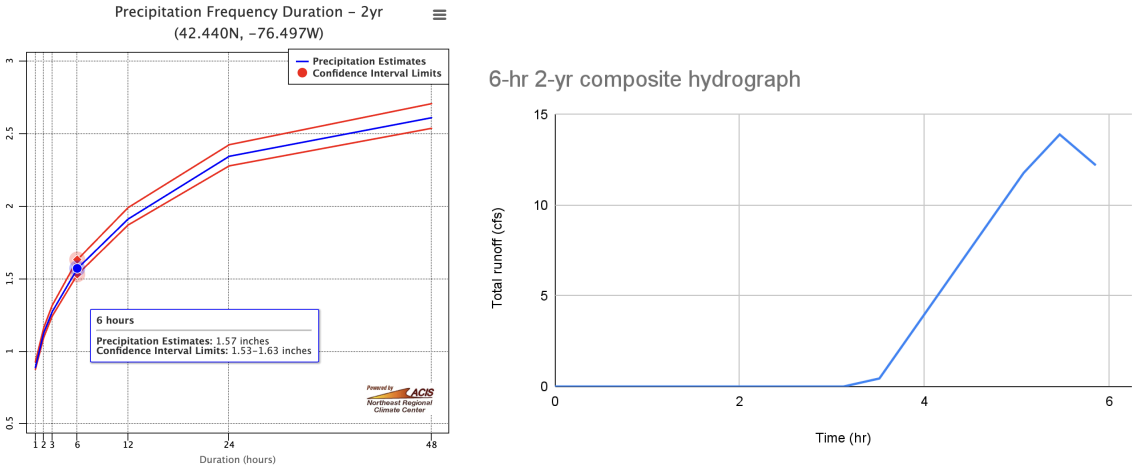


Figure B3. NRCC extreme precipitation graph for the 6-hr 2-yr storm in Ithaca, New York and the composite hydrograph generated by runoff from the storm. Runoff begins after 3.5 hours.

Then, this same process was repeated to determine the frequency the park would flood by comparing the maximum flow rate generated by the storm with the maximum flow rate the new channel can handle. The maximum flow rate the channel can handle is found using Manning's equation. The channel is concrete lined, with a design Manning's number *n* of 0.015, slope of 0.0026, and dimensions of 29 ft wide and 2.5 ft high on one side and 3 ft 10 inch on the other. Since the channel is limited by the shorter wall, 2.5 ft was used for all channel heights. The hydraulic radius is calculated using the following equation:

$$R = A/WP = (348 * 30)/(348 + 2 * 30) = 25.59\,in\,or\,0.65\,m \quad (1)$$

With the hydraulic radius, the velocity can be calculated using Manning's equation shown in Equation B2. Finally, the maximum stream flow rate is found by multiplying velocity by the cross sectional area of the channel, and equals approximately 606.8 cfs for the existing channel. There is a 50% chance of a storm that could cause flooding each year.

$$v = (1/n)R^{2/3}S^{1/2} = (1/0.015)(0.65)^{2/3}(0.0026)^{1/2} = 2.55\,m/s\,or\,8.37\,ft/s \quad (2)$$

$$Q = v * A = (8.37)(29 * 2.5) = 606.83\,cfs \quad (3)$$

This approach was repeated with the new channel area, which is 29 ft^2 since the wall height will be 1 ft. Using Equations B1 through B3 again, the redesigned channel maximum flow rate will be approximately 145 cfs. Several storms were found for the comparison of maximum storm versus channel flow. The maximum flow of a storm is determined by the largest sum of unit hydrograph flow rates at any given time. An example for the 12-hr 1-yr storm is shown in Table B1.

Table B1. Example of the flow rates generated by the 12-hr 1-yr storm at given times during the storm. The overall maximum flow rate for the entire storm is the largest value, 120.26 cfs.

time (hr)	q Total (cfs)
4.29	0.41
4.68	0.44
5.07	1.39
5.46	4.48
5.85	10.88
6.24	21.03
6.63	37.23
7.02	58.40
7.41	83.52
7.80	103.36
8.19	116.17
8.58	120.26
8.97	114.06
9.36	96.21
9.75	78.30
10.14	60.67
10.53	44.07
10.92	29.20

11.31	16.74
11.70	7.32
12.09	1.55

The channel was found to completely handle the flows of a 12-hr 1-yr storm (maximum flow rate of 120.26 cfs) or 6-hr 2-yr storm (maximum flow rate of 112 cfs).

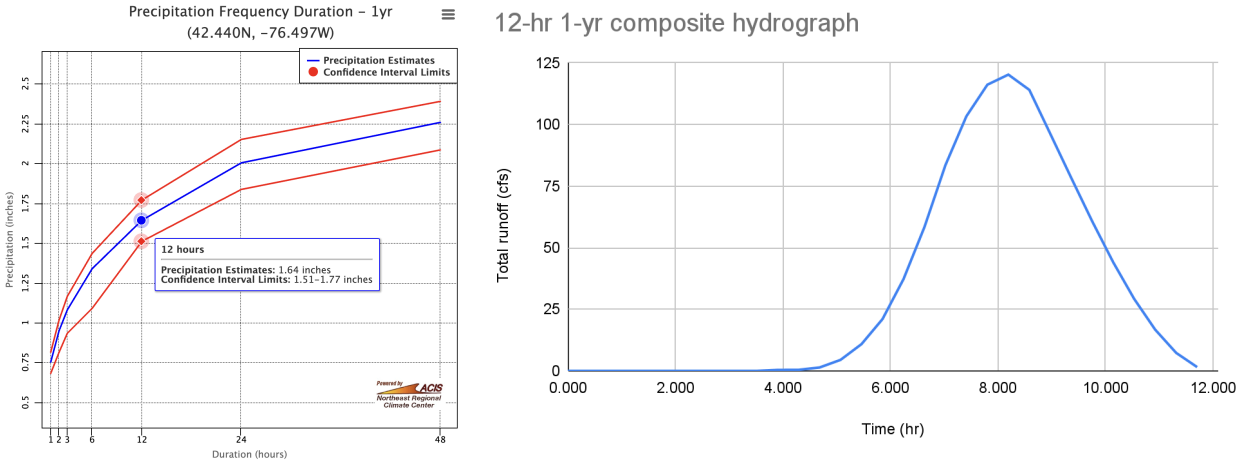


Figure B4. NRCC extreme precipitation graph for the 12-hr 1-yr storm in Ithaca, New York, and the corresponding composite hydrograph. The maximum runoff flow rate is 120.26 cfs.

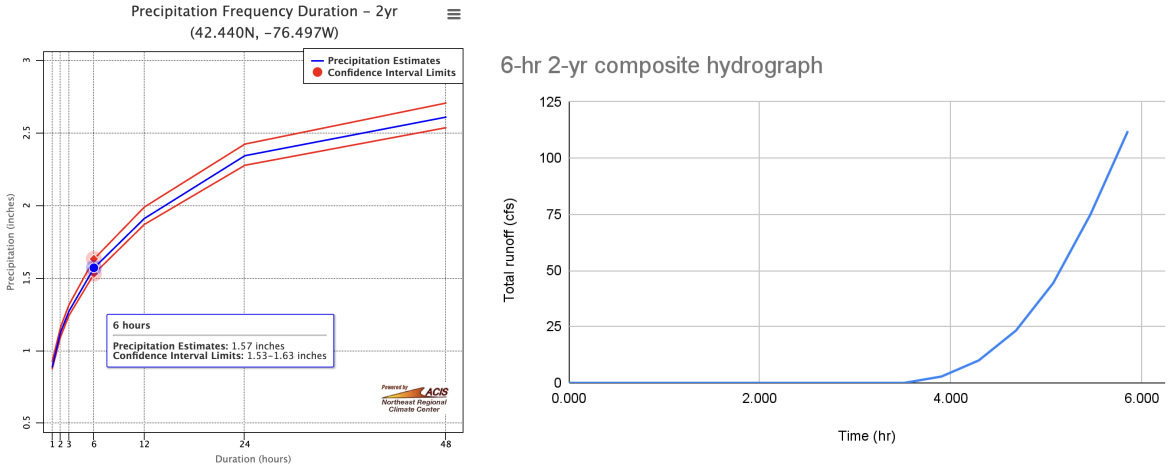


Figure B5. NRCC extreme precipitation graph for the 6-hr 2-yr storm in Ithaca, New York, and the corresponding composite hydrograph. The maximum runoff flow rate is 112 cfs.